

Module - 2

Random Variable - Probability mass function, distribution and density function - Joint probability distribution and joint density function, Marginal, conditional distribution and Density functions - Mathematical expectation and its properties - covariance, Moment generating function.

Random Experiment:-

If in each trial of an experiment conducted under identical conditions, the outcome is not unique but may be one of the possible outcomes, then such an experiment is called random experiment.

ex - tossing of a coin, throwing a die, selecting a card from a pack of card

Trial and event:-

Any particular performance of a random experiment is called a trial.

Outcome or combination of outcome are called events.

For ex:- If a coin is tossed repeatedly, we may get any one of the two faces. head or tail.

i.e. Tossing of a coin is random experiment or trial.

Getting head or tail is an event.

or (cases)

Exhaustive events:- The total no. of possible outcome of a random experiment is known as exhaustive events.

Ex 1) In tossing a coin, there are two exhaustive events, head & tail.

2) In throwing a dice, there are 6 exhaustive events 1, 2, 3, 4, 5, 6

(3) In throwing two dice, then exhaustive events are $6^2 = 36$

2) Mutually exclusive events:- Events are said to be mutually exclusive if happening of any one precludes the happening of other.

(1) In throwing a dice, all the 6 faces numbered 1 to 6 are mutually exclusive, since if any one of the faces comes the possibility of other in same trial is ruled out.

(2) In tossing of a coin, events head and tail are mutually exclusive.

Independent Events:- Events are said to be independent if the happening of an event is not affected by other events.

Ex when a die is thrown twice, the result of the first throw does not affect the result of second throw.

Ex:- If we draw a card from a pack of well shuffled cards & replace it before drawing the second card, the result of second draw is independent of first draw. But, if the first drawn card is not replaced, then the second draw is dependent of first draw.

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Probability

Random variable:-

Sample Space:- A sample space is the collection or a set of all possible outcomes of a random experiment. Sample space is represented by S .

Ex • For rolling a dice, sample space is

$$S = \{1, 2, 3, 4, 5, 6\}$$

• When flipping a coin, sample space is

$$S = \{\text{Head}, \text{Tail}\}$$

• Tossing of two coins, the possible outcomes are 4 i.e. $S = \{HH, HT, TH, TT\}$ for n coins, possible outcome 2^n

Random Variable:-

The outcomes of a random experiments may be numerical or non-numerical in nature.

To describe the outcome of a random experiment by a number, each non-numerical outcome of the experiment is assigned a number. Thus

Random variable is a function that assigns a real number to each element in the sample space.

$$\text{i.e. } X: S \rightarrow R$$

such that $X(s) = x$, where $s \in S$

Ex Flipping a coin $S = \{\text{Head}, \text{Tails}\}$

let give them values

$$\text{Head} = 0, \text{Tail} = 1$$

Random variable

$$X = \{0, 1\}$$

Ex

Tossing of two coins, $S = \{HH, HT, TH, TT\}$
define a random variable X which is
no. of heads:

outcome:	TT	TH	HT	HH
no. of heads	0	1	1	2

Here we are interested in no. of head. ^{ie} we define a random variable x , whose ^{classmate} value ^{Date} is no. of heads ^{Page}
 $ie\ x = \{0, 1, 2\}$, Sample Space = $\{TT, TH, HT, HH\}$

^{ie} In a Random experiment, give value to each event, The set of values is Random Variable.

Now Probability of each event is define as
 $P(X = \text{value}) = \text{Probability of that value}$

^{ie} in above example

$P(X=0)$ is Probability of getting 0 head
^{ie}

$$P(X=0) = \frac{1}{4}, \quad P(X=1) = \frac{2}{4}, \quad P(X=2) = \frac{1}{4}.$$

Q How many head when three coins are tossed.
Sample Space = $\{HHH, HTH, THH, TTH, HHT, HTT, THT, TTT\}$
 $X = \{0, 1, 2, 3\}$

$$P(X=0) = \frac{1}{8}, \quad P(X=1) = \frac{3}{8}, \quad P(X=2) = \frac{3}{8}, \quad P(X=3) = \frac{1}{8}$$

Note:- Total Probability is always 1

P of an event is $P(E) \geq 0$ & $P(E) \leq 1$

① $ie\ 0 \leq P(E) \leq 1$

② $P(E) + P(\bar{E}) = 1$

Discrete random Variable:-

A random variable x is said to be discrete if it can take a finite number or countably infinite no. of value. When the Random variable is discrete, the possible value of x be assumed as $x_1, x_2, \dots, x_n$.

Ex Example of discrete random variables are marks obtained in a test, the no. shown when dice is thrown, no. of telephone calls per month, no. of α particles emitted by radioactive source.

Probability Mass function:- (P.m.f)

If x is discrete random variable which takes the values $x_1, x_2, \dots$ then its probabilistic behaviour at each point is defined by a function called probability mass function, given by (or discrete density function)

$$p(x) = P(X = x_i) \\ = p_i \text{ or } p(x_i)$$

or we write

$$\boxed{P(X = x_i) = p_i} \quad i = 1, 2, 3, \dots$$

The $p_i, i = 1, 2, 3, \dots$ satisfy the following conditions:-

(i) $p_i \geq 0$

and (ii) $\sum_i p_i = 1$.

The collection pair $\{(x_i, p_i), i = 1, 2, \dots\}$ or $\{(x_1, p_1), (x_2, p_2), \dots, (x_n, p_n), \dots\}$ is called probability distribution of r.v. X .

which is sometime displayed in the form of table

$X = x_i$	$P(X = x_i)$
x_1	P_1
x_2	P_2
$\vdots$	$\vdots$
x_n	P_n
$\vdots$	$\vdots$

Q A die is tossed twice. Getting 'an odd no.' is termed as success. Find the probability distribution of the no. of Success.

Solⁿ:- In one throw, Probability of getting odd no, say termed as success then

$$P(\text{success}) = \frac{3}{6} = \frac{1}{2} \quad \text{ie } P(S) = \frac{1}{2}$$

$$P(\text{Failure}) = \frac{3}{6} = \frac{1}{2} \quad P(F) = \frac{1}{2}$$

Let X be a random variable, which denotes the success in two throw. then X take the values

$$X = \{0, 1, 2\}$$

then

$$P(X=0) = P(F \text{ in first throw and } F \text{ in second throw})$$

$$= P(F) \times P(F)$$

$$P(X=0) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$\boxed{P(X=0) = \frac{1}{4}}$$

Using

$P(A \text{ and } B)$

$$= P(A) \cdot P(B)$$

Prob. of

Independent
even

$$\begin{aligned}
 P(X=1) &= P(S \text{ and } F) + P(F \text{ and } S) \\
 &= P(S) \cdot P(F) + P(F) \cdot P(S) \\
 &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}
 \end{aligned}$$

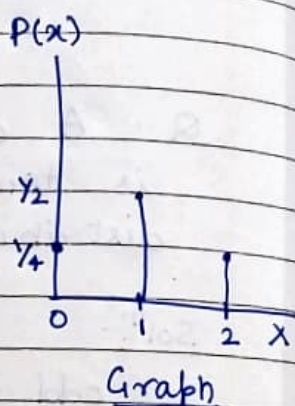
$$[P(X=1) = \frac{1}{2}]$$

$$\begin{aligned}
 P(X=2) &= P(S \text{ and } S) \\
 &= P(S) \cdot P(S) \\
 &= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}
 \end{aligned}$$

$$[P(X=2) = \frac{1}{4}]$$

Probability function of X

$X :$	0	1	2
$P(x) :$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$



Q2

Two cards are drawn

- (a) Successively with replacement
(b) Simultaneously (without replacement)
- from a well shuffled deck of 52 cards.
Find the probability distribution of the no. of aces.

Solⁿ

Let X is a random variable, denotes the no. of aces in draw of two cards.

then $X = \{0, 1, 2\}$

$P(\text{aces}) = \dots$

(a) without replacement:-

$$P(\text{ace}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{non-ace}) = 1 - \frac{1}{13} = \frac{12}{13}$$

Now

$$\begin{aligned}
 P(X=0) &= P(\text{non-ace and non-ace}) \\
 &= P(\text{nonace}) \times P(\text{non-ace}) \\
 &= \frac{12 \times 12}{13 \times 13} = \frac{144}{169}
 \end{aligned}$$

$$P(X=0) = \frac{144}{169}$$

$$\begin{aligned}
 P(X=1) &= P(\text{ace and Nonace}) + P(\text{Non-ace and ace}) \\
 &= P(\text{ace}) \cdot P(\text{nonace}) + P(\text{nonace}) \cdot P(\text{ace}) \\
 &= \frac{1}{13} \cdot \frac{12}{13} + \frac{12}{13} \cdot \frac{1}{13} \\
 &= \frac{24}{169}
 \end{aligned}$$

$$P(X=1) = \frac{24}{169}$$

$$\begin{aligned}
 P(X=2) &= P(\text{ace and ace}) \\
 &= P(\text{ace}) \cdot P(\text{ace}) = \frac{1}{13} \cdot \frac{1}{13}
 \end{aligned}$$

$$P(X=2) = \frac{1}{169}$$

~~X=2~~

X :	0	1	2
P(X) :	$\frac{144}{169}$	$\frac{24}{169}$	$\frac{1}{169}$

(b) without replacement:

$$P(\text{ace in first draw}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{Non ace in first draw}) = \frac{12}{13}$$

Let P_1 = Prob of getting ace in second draw, if first drawn card is ace P_2 = Prob of getting ace in second draw if first drawn card is nonace

$$P_1 = \frac{3}{51} \quad P_1^c = 1 - \frac{3}{51} = \frac{48}{51}$$

$$P_2 = \frac{4}{51} \quad P_2^c = 1 - \frac{4}{51} = \frac{47}{51}$$

Now $P(X=0) = P(\text{Nonace and Nonace})$
 $= P(\text{Nonace}) \times P_2(\text{nonace})$
 $= \frac{12}{13} \times \frac{47}{51} = \frac{188}{221}$

$$P(X=1) = P(\text{ace}) \cdot P_1^c + P(\text{nonace}) \cdot P_2$$

$$= \frac{12}{13} \times \frac{48}{51} + \frac{12}{13} \times \frac{4}{51}$$

$$[P(X=1) = \frac{32}{221}]$$

Now $P(X=2) = P(\text{nonace and nonace})$
 $= P(\text{nonace}) \times P_2^c$

$$P(X=2) = P(\text{ace and ace})$$

$$= P(\text{ace}) \times P_1$$

$$= \frac{4}{52} \times \frac{31}{51} = \frac{1}{221}$$

Probability distribution

$X=0$	1	2
$P(X) = \frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$

Q Obtain the probability distribution of X , the no. of head in three tosses of a coin.

Solⁿ: Sample space = $\{(H, T) \times (H, T) \times (H, T)\}$
 $= \{(HH, HT, TH, TT) \times (H, T)\}$
 $= \{HHH, HTH, THH, TTH, HHT, HTT, THT, TTT\}$

X is a random variable which denotes no. of Heads in three tosses of a coin.

i.e. $X = \{0, 1, 2, 3\}$

ie $P(X=0) = \frac{1}{8}$

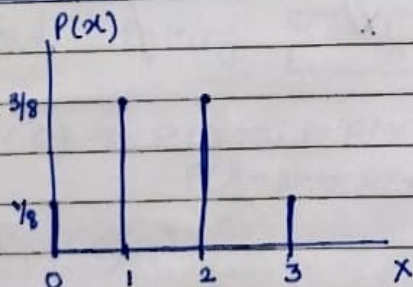
$P(X=1) = \frac{3}{8}$

$P(X=2) = \frac{3}{8}$

$P(X=3) = \frac{1}{8}$

i.e. Probability distribution is

$X:$	0	1	2	3
$P(X):$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$



Q:- If the random variable takes 4 values 1, 2, 3, 4 such that

$2 P(X=1) = 3 P(X=2) = P(X=3) = 5 P(X=4)$, find the probability distribution function of X .

Solⁿ:- Say $2 P(X=1) = 3 P(X=2) = P(X=3) = 5 P(X=4) = k$

then $P(X=1) = \frac{k}{2}$, $P(X=2) = \frac{k}{3}$, $P(X=3) = k$,
 $P(X=4) = \frac{k}{5}$

we know that $\sum P_i = 1$

$$\text{i.e. } \frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1$$

$$k = \frac{30}{61}$$

i.e. Probability distribution of X is given by

X_i	1	2	3	4
$P(X_i)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$

commulative distribution function:

The commulative distribution function is define as $P(X \leq x)$

i.e. $P(X < 1) = 0$

$P(X)$

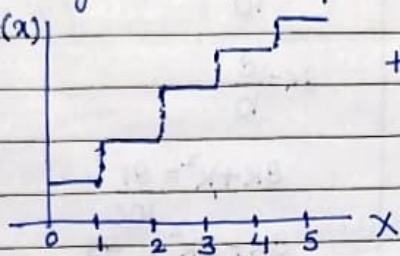
Cumulative distribution function:- (C.d.f) (for discrete variable)

Let X be a discrete random variable, then the function

$$F(x) = P(X \leq x)$$

is called cumulative distribution function of X . $F(x)$ is a step function having jump p_i at $x = x_i$.

$F(x)$



Say $x_1, x_2, \dots, x_n$ are pts then $P(X \leq x_i) = \sum_{j=1}^i P(x_j)$

Q:- A random variable X has the following probability function

x :	0	1	2	3	4	5	6	7
$P(x)$	0	K	$2K$	$2K$	$3K$	K^2	$2K^2$	$7K^2 + K$

(i) Find K (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$

(iii) $P(X \leq a) > \frac{1}{2}$, find the minimum value of a and (iv) Determine the ^{commulative} distribution function of X .

solⁿ (i) Since total probability = 1

$$\sum_{i=0}^7 P(x_i) = 1$$

$$\text{ie } 10K^2 + 9K - 1 = 0 \text{ ie } K = \frac{1}{10} \text{ or } K = -1$$

Since $P(x)$ cannot be negative ie $K = -1$ is rejected. taking $K = \frac{1}{10}$

$$(ii) P(X < 6) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$P(X < 6) = \frac{81}{100}$$

$$P(X \geq 6) = 1 - P(X < 6) = \frac{19}{100}$$

distribution function of X

X	$F(x) = P(X \leq x)$
0	0
1	$k = \frac{1}{10}$
2	$3k = \frac{3}{10}$
3	$5k = \frac{5}{10}$
4	$8k = \frac{8}{10}$
5	$8k + k^2 = \frac{81}{100}$
6	$8k + 3k^2 = \frac{83}{100}$
7	$9k + 10k^2 = 1$

Now $P(X \leq a) \geq \frac{1}{2}$

from table its clear that

$$\boxed{a=4}$$

$$\text{as } P(X \leq 4) = .8 > \frac{1}{2}$$

Q. If $P(x) = \begin{cases} \frac{x}{15}, & x=1, 2, 3, 4, 5 \\ 0 & \text{e.w.} \end{cases}$

Find (i) $P(X=1 \text{ or } 2)$ (ii) $P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right)$ Solⁿ: (i) $P(X=1 \text{ or } 2)$

$$= P(X=1) + P(X=2)$$

$$= \frac{1}{15} + \frac{2}{15} = \frac{1}{5}$$

(ii) $P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right)$

$$= \frac{P\left(\frac{1}{2} < X < \frac{5}{2}\right) \cap (X > 1)}{P(X > 1)}$$

Conditional Prob.

$P(A/B)$ is conditional Prob ie probability of event A, when B is already occurred

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Similarly

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(X > 1) = 1 - P(X \leq 1)$$

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$$P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) = \frac{P(X=1 \text{ or } 2) \cap P(X > 1)}{P(X > 1)}$$

$$= \frac{P(X=2)}{P(X > 1)}$$

$$= \frac{2/15}{1 - P(X=1)} = \frac{2/15}{1 - 1/15} = \frac{1}{7}$$

$$P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) = \frac{1}{7}$$

Continuous Random Variable:-

If X is an random variable, which take all the values (infinite no.) in an interval, then X is called a continuous random variable. examples of continuous random variables are age, height, weight.

Probability density function:- (p.d.f)

The probability density function of a ^{cont.} random variable X , denoted by $f_X(x)$ or $f(x)$ satisfying the following conditions:-

(i) $f(x) \geq 0 \quad \forall \quad x \in R_X$

(ii) $\int_{R_X} f(x) dx = 1$

$$P\left(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2}\right) = f(x) dx$$

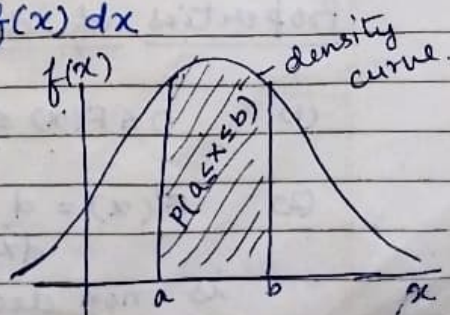
then $f(x)$ is called P.d.f.

(iii) The probability of any event E is given by $P(E)$

$$P(E) = \int_E f(x) dx$$

[$P(E)$ is the area under the curve.]

Area b/w the density curve and x axis is



Note:- For discrete random variable, the probability at a point $P(X=a)$ is not zero, but for continuous random variable, the probability at a point $X=a$ is defined as

$$P(X=a) = P(a \leq X \leq a) = \int_a^a f(x) dx = 0$$

ie it is always zero, ^a at a point.
ie $P(X=a) = 0$ for cont. random variable.
This prop of cont. random variable leads to following result

$$\begin{aligned} * \quad P(a \leq X \leq b) &= P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) \\ &= \int_a^b f(x) dx, \quad X \text{ is cont. r.v.} \end{aligned}$$

ie In case of cont. random variable it does not matter whether we include end points of interval.

Commutative Distribution function:- (C.d.f)
(for continuous variable)

If X is ~~discrete~~ continuous random variable, the $P(X \leq x)$ is called commulative distribution function of X , denoted by $F(x)$

given by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

Properties of distribution function:-

$$(1) \quad 0 \leq F(x) \leq 1$$

$$(2) \quad F'(x) = \frac{d}{dx} F(x) = f(x) \geq 0 \text{ ie } F(x)$$

is non decreasing function of x .

ie $F(x_1) \leq F(x_2)$ if $x_1 < x_2$.

$$(3) \quad F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} \int_{-\infty}^x f(x) dx = \int_{-\infty}^{-\infty} f(x) dx$$

$$F(-\infty) = 0$$

$$F(+\infty) = \lim_{x \rightarrow +\infty} F(x) = \int_{-\infty}^{+\infty} f(x) dx = 1$$

$$\text{ie } F(+\infty) = 1$$

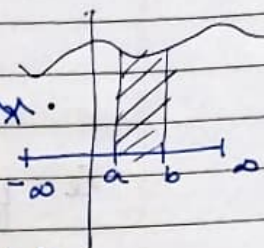
$$\text{ie } 0 \leq F(x) \leq 1$$

$$(4) \quad P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area}$$

$$= \int_{-\infty}^b f(x) dx - \int_{-\infty}^a f(x) dx$$

$$= P(X \leq b) - P(X \leq a)$$

$$\boxed{P(a \leq X \leq b) = F(b) - F(a)}$$



$$(5) \quad \boxed{\frac{d}{dx} F(x) = f(x)}$$

ie we can obtain p.d.f from distribution ~~den~~ function $F(x)$, by differentiating.

Ex 1 - $f(x) = 6x(1-x)$, $0 \leq x \leq 1$.
check $f(x)$ is probability density function (p.d.f)

Soln (i) $f(x) = 6x(1-x) \geq 0$ $\forall$ $0 \leq x \leq 1$.

$$(ii) \int_0^1 f(x) dx = \int_0^1 6x(1-x) dx$$

$$= 6 \left(\frac{x^2}{2} \right)_0^1 - 6 \left(\frac{x^3}{3} \right)_0^1$$

$$\int_0^1 f(x) dx = 1$$

$\therefore f(x)$ is p.d.f.

Ex 2 Find the value of k for the p.d.f.

$$f(x) = \begin{cases} kx^2, & 0 \leq x \leq 3 \\ 0 & \text{o.w} \end{cases}$$

also compute $P(1 \leq x \leq 2)$ and distribution function.

Soln for $f(x)$ to be p.d.f

$$\int_0^3 f(x) dx = 1$$

$$\int_0^3 kx^2 dx = 1$$

$$\boxed{k = 1/9}$$

Now

$$P(1 \leq x \leq 2) = \int_1^2 f(x) dx = \frac{7}{27}$$

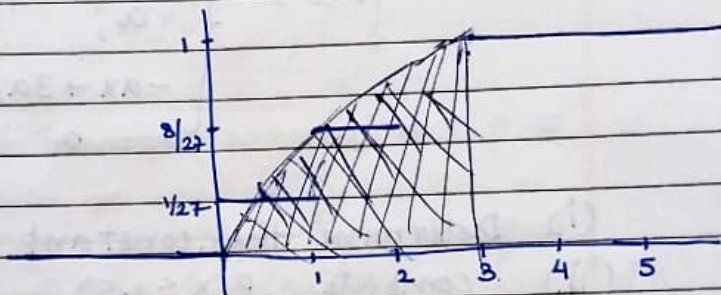
Now, for distribution function $F(x)$

$$F(x) = \int_{-\infty}^x f(x) dx$$

$$= \int_0^x f(x) dx = \int_0^x \frac{1}{9} x^2 dx$$

$$F(x) = \frac{x^3}{27}$$

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^3}{27}, & 0 \leq x \leq 3 \\ 1, & x > 3 \end{cases}$$



- Q. A continuous random variable X has P.d.f $f(x) = 3x^2$, $0 \leq x \leq 1$. Find a and b such that
- (i) $P(X \leq a) = P(X > a)$ (ii) $P(X > b) = 0.05$

Soln:- P.d.f $f(x) = 3x^2$, $0 \leq x \leq 1$

given $P(X \leq a) = P(X > a)$

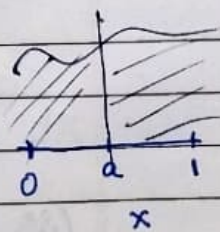
$$\int_0^a f(x) dx = \int_a^1 f(x) dx$$

$$\int_0^a 3x^2 dx = \int_a^1 3x^2 dx$$

$$3 \left(\frac{x^3}{3} \right)_0^a = 3 \left(\frac{x^3}{3} \right)_a^1$$

$$a^3 = 1 - a^3$$

$$2a^3 = 1 \Rightarrow a = \left(\frac{1}{2} \right)^{\frac{1}{3}}$$



$$(11) \quad P(X > b) = 0.05$$

$$\int_b^1 f(x) dx = 0.05$$

$$3 \left(\frac{x^3}{3} \right)_b^1 = 0.05$$

$$1 - b^3 = 0.05$$

$$1 - 0.05 = b^3$$

$$b = \sqrt[3]{0.95}$$

$$b = \left(\frac{19}{20} \right)^{1/3}$$

Ans

Q Let X be a continuous random variable with p.d.f

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 \leq x \leq 3 \\ 0 & \text{e.w.} \end{cases}$$

(i) Determine the constant a

(ii) compute $P(X \leq 1.5)$

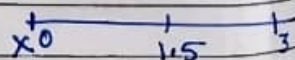
(iii) find c.d.f

Solⁿ (i) Using $\int_{-\infty}^{\infty} f(x) dx = 1$ i.e. $\int_0^3 f(x) dx = 1$

$$\text{i.e. } \int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (-ax + 3a) dx = 1$$

$$a = \frac{1}{2}$$

$$(ii) \quad P(X \leq 1.5) = \int_0^{1.5} f(x) dx$$



$$= \int_0^{1.5} f(x) dx = \int_0^1 ax dx + \int_1^{1.5} a dx$$

$$= \frac{a}{2} + 1.5a$$

$$= a$$

$$= \frac{1}{2}$$

ie

$$P(X \leq 1.5) = \frac{1}{2}$$

c.d.f is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

$$= \int_{-\infty}^x f(x) dx$$

$$= \int_0^1 x dx + \int_1^2 \frac{1}{2} dx + \int_2^3 0 dx$$

Now for $-\infty < x < 0$, $F(x) = \int_{-\infty}^x 0 dx = 0$

for $0 \leq x < 1$, $F(x) = \int_{-\infty}^x f(x) dx = \int_0^x \frac{x^2}{2} dx$

$$0 \leq x < 1, \quad F(x) = \frac{x^3}{6}$$

For, $1 \leq x < 2$, $F(x) = \int_{-\infty}^x f(x) dx$

$$= \int_0^1 \frac{x^2}{2} dx + \int_1^x \frac{1}{2} dx$$

for $1 \leq x < 2$, $F(x) = \frac{x^2 - 1}{4}$

For $2 \leq x \leq 3$

$$F(x) = \int_0^1 \frac{x^2}{2} dx + \int_1^2 \frac{1}{2} dx + \int_2^x 0 dx$$

$$2 \leq x \leq 3, \quad F(x) = -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4}$$

For any x , $3 \leq x < \infty$

$$\begin{aligned} F(x) &= \int_{-\infty}^0 0 \, dx + \int_0^1 \frac{x}{2} \, dx + \int_1^2 \frac{1}{2} \, dx + \int_2^3 \left(-\frac{x}{2} + \frac{3}{2}\right) dx \\ &\quad + \int_3^{\infty} 0 \, dx \\ &= \frac{1}{4} + \left(1 - \frac{1}{2}\right) + \left(-\frac{9}{4} + \frac{9}{2} + 1 - 3\right) = 1 \end{aligned}$$

$$\text{ie } F(x) = \begin{cases} 0, & -\infty \leq x < 0 \\ \frac{x^2}{4}, & 0 \leq x < 1 \\ \frac{2x-1}{4}, & 1 \leq x < 2 \\ -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4}, & 2 \leq x < 3 \\ 1, & 3 \leq x < \infty. \end{cases}$$

- Q A petrol pump is supplied with petrol one a day. If its daily volume of sales (x) in thousand of liters is distributed by $f(x) = 5(1-x)^4$, $0 \leq x \leq 1$. What must be the capacity of tank in order that the probability its supply will be exhausted in a given day shall be 0.01.

Soln. let capacity of tank be a litre.

then $P(X \geq a) = 0.01$

$$\int_a^1 f(x) \, dx = 0.01$$

$$\int_a^1 5(1-x)^4 \, dx = 0.01$$

$$(1-a)^5 = \frac{1}{100} \Rightarrow a = 1 - \left(\frac{1}{100}\right)^{1/5}$$

$$= 1 - 0.3981$$

$$= 0.6019$$

$$\text{Capacity of tank} = 0.6019 \times 1000 = 601.9 \text{ lit.}$$

Suppose life in hours of a certain part of radio tube is continuous random variable X with p.d.f given by

$$f(x) = \begin{cases} \frac{100}{x^2}, & x \geq 100 \\ 0 & \text{e.o.w} \end{cases}$$

- (i) what is the probability that all of three such tube in a given radio set will have to be replaced during the first 150 hours of operation?
- (ii) What is the probability that none of three of the original tubes will have to be replaced during that first 150 hours of operation?
- (iii) what is the probability that a tube will last less than 200 hours if it is known that the tube is still functioning after 150 hours of service.

Soln:- (i) $P(X \leq 150) = \int_{100}^{150} \frac{100}{x^2} dx = \frac{1}{3}$

ie Prob that a tube is replaced during first 150 hours = $\frac{1}{3}$

ie Prob that all three of original tube is replace during 150 hours = $\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \left(\frac{1}{3}\right)^3$

(ii) $P(X > 150) = 1 - P(X \leq 150) = 1 - \frac{1}{3} = \frac{2}{3}$

Prob that none of the three tube will be replace during first 150 hours = $\left(\frac{2}{3}\right)^3$

(iii) Prob that tube will last less than 200 hours such that it is still functioning after 150 hours of operation

$$\begin{aligned} P(X < 200 | X > 150) &= \frac{P(X < 200 \cap X > 150)}{P(X > 150)} \\ &= \frac{P(150 < X < 200)}{P(X > 150)} = 0.25 \end{aligned}$$